Practice Assignment 3

Bayesian Methods and Modeling

Discussion on Monday, May 11th at 4 PM

1) Suppose we have a data set with each observation x_i is independent and identically exponentially distributed for $i = 1,2, \dots, n$. That is, $x_i \sim Exp(\lambda)$ where λ is the rate parameter. We would like to find a posterior (or at least a function proportional to it) for λ.
    a. Write down the likelihood function in this situation. We would call this $p(\boldsymbol{x}|\lambda)$.

    b. Now let λ have a normal prior with mean 0.1 and variance 1: $\lambda \sim N(0.1, 1)$. Use this and the likelihood from part (a) to write down a function that is proportional to the posterior of λ given $\boldsymbol{x}$. We call this $p(\lambda|\boldsymbol{x})$. Remember that any term without λ that is being multiplied can be ignored.

2) Recall the **Country_Information.csv** dataset in the Practice 1 folder on GitHub. In this dataset, some variables were collected from 41 countries around the world. The variables in the dataset are
    1. Life expectancy (LifeExpectancy),
    2. Average number of years of schooling (YearsSchool),
    3. Gross domestic product per capita (GDP) in U.S. dollars,
    4. The gender inequality index (GII), which rates the amount of gender inequality in the given country (bigger values mean more inequality),
    5. An index score rating the quality of education in the given country (EduIndex), and
    6. Public health expenditures, which measures how much the given country spends on public health programs (PubHealthExp).

    a. Read the dataset into **R**. Also, if you have not already, install the **math3150package** library from my GitHub (instructions for this are in the notes) and load that package. This will also load the tidyverse.

    b. Fit a linear regression model using GII as the explanatory variable and life expectancy as the response variable like you did in the Practice 1 assignment. Write down the model in the context of the problem.

    c. Now use the **brm()** function in the **brms R** package to fit a Bayesian model predicting life expectancy using GII. Use the default priors and the "cmdstanr" backend. Write down the model using the Estimate values.

    d. Compare the model estimates of the intercept, slope, and sigma value for the linear regression model fit in part (b) and the Bayesian model fit in part (c). How similar or different are the values?

    e. Compare the standard error and estimated errors for the intercept and slope for the linear regression and Bayesian models. How similar or different are the values?

f. Plot the Bayesian model to see the sampling chains for b_Intercept, b_GII, and sigma. Comment on whether the chains look good. Also comment on the effective sample sizes and R-hat values for the three parameters.

g. Not put a normal prior on the intercept with a mean of 50 and a standard deviation of 10, put a normal prior on the slope of GII with a mean of -20 and a standard deviation of 5, and put an exponential prior on sigma with a rate parameter of 0.2 (you can set that last prior for sigma using **prior(exponential(0.2), class = "sigma")**). Fit a Bayesian model with those priors and again using the "cmdstanr" backend. Comment on the differences in the estimates between this model and the previous Bayesian model. Do those differences make sense?

h. Now use the **pp_check()** function to plot some replications of y. Comment on how it looks.

3) Now again consider the **USTempLocation.csv** dataset from Practice 2.

a. Read in the dataset, change CitySize to a factor with the levels "Under1M" and "Over1M" like you did in Practice 2, and then fit a logistic regression model predicting CitySize from JanTemp and Region. Write down the model here.

b. Now fit a Bayesian logistic regression model using **brm()** with **family = bernoulli()** with the default priors and using "cmdstandr" as the backend. Write down this model.

c. In the logistic regression model, only JanTemp (and the intercept) was considered significant since its p-value was less than 0.05. For Bayesian models, we can tell which variables are significant based on whether zero is in the Bayesian credible interval or not. If both bounds are positive or both negative, then the variable is significant. If one bound is positive and one is negative, then it is not significant. Based on the l-95% CI and u-95% CI values in the output, which variables are considered significant in the Bayesian model?

d. Now set the following priors:
For JanTemp: normal with mean 1 and SD 1.
For RegionNortheast: normal with mean 2 and SD 1.
For RegionSouth: normal with mean -2 and SD 1.
For RegionWest: normal with mean 0 and SD 1.

Then fit a Bayesian model with "cmdstanr" backend, 4000 iterations, 6 chains, 4 cores, and a seed of 1. Type **?brm** in the R console for more info on how to set iterations, chains, cores, and the seed. Setting the cores to 4 will run 4 chains at the same time in parallel across different cores on your computer.

Looking at the output of the new model, which variables are now considered significant?

e. The **predict()** function also works with Bayesian models. However, it gives more information than for a linear or logistic model. The predict estimates are in the first column of the output of **predict()**. So, you could type **predict(bayesian_model)[,1]** to get the predictions. Produce a table with all the actual and all the predicted values from the model with the custom priors you fit in part (d) using 0.50 as the probability cutoff for predicted the city has over 1 million people.

What percent of cities were classified correctly?

f. Use the **pp_check()** function on this Bayesian model with the custom priors and comment on it.